Disease Detectives

Answer Key

1. 6-15 hours (2 points)
2. Contact, vehicle, vector (1 point)
3. a) >/= 80 (1 point)
   b) All ages (1 point)
   c) Accepted answer in range: 47% to 53% (1 point)
4. a) 10/20 (1 point)
   b) 10/17 to 11/21 (1 point)
5. a) 0.8 (2 points)
   b) 0.45 (2 points)
   c) 1.78 (+ or - 0.02) (2 points)
   d) Increasing exposure to salad is associated with increasing risk of developing gastroenteritis. (2 points)
   e) cohort study (1 point)
   f) diarrhea, cramps, nausea, fever, vomiting (2 points)
6. a) 3.52 (+ or - 0.02) (2 points)
   b) This shows that people who ate at Chipotle are 3.52 times more likely to develop the disease than those who did not eat at Chipotle. (2 points)
   c) Case control (1 point)
   d) Retrospective (1 point)
   e) Cochran-Mantel-Haenszel summary odds ratio/ Cochran-Mantel-Haenszel estimate of the risk ratio and of the odds ratio (1 point)
7. Line listing (1 point)
8. 4 (1 point)
9. Discrete onset of symptoms, presence of jaundice, and a positive serological test for IgM antibody to Hepatitis A (1 point)
10. 6 (1 point)
11. Immunoglobulin G (1 point)
12. It is an antibody test that detects the first antibody that the body produces when it is exposed to hepatitis A. It is used to diagnose acute hepatitis A. (2 points)

13. a) The temperature of the fridge was taken to indicate whether it was cool enough to allow any bacteria in the food to grow. (2 points)

    b) The temperature of the fat was taken to indicate whether it was hot enough to kill any bacteria present in the food. (2 points)

14-16: (2 points for each correct answer, 1 point given for identifying a correct risk factor but not a correct explanation) Any three of the following are correct:

Cracked work surface- Bacteria can grow in the cracks of the work surface.

Meat grinder has putrefied meat adhering to it- Bacteria from the putrefying meat may spread to other foods.

No convenient hand washing facilities- Hand washing is an important precaution to take to prevent the spread of disease. If there are no convenient hand washing facilities, this cannot be done and disease can spread much more quickly.

Temperature of the fridge- The fridge temperature measured 68 degrees, which is within the range of temperatures that allow bacteria to proliferate (40 degrees to 140 degrees F).

17. A coliform test measures the concentration of total coliform bacteria associated with the possible presence of disease causing organisms. It indicates the suitability for consumption, or potability, of drinking water. (2 points)
18. *E. coli* (1 point)
19. Cohort study (1 point)
20. 0.372 or 37.2% (+ or - 0.002 or 0.2%) (1 point)

Matching Answers: (1 point each)

1. I
2. A
3. J
4. V
5. P
6. Q
7. K
8. D
9. E

10. G
11. H
12. M
13. F
14. L
15. S
16. T
17. R
18. C
19. B
20. N
21. O
22. U

Max Total: 70

Tie Breaker Answers:

1. Random error- the divergence of an observed sample value from the true population value due to chance alone (1 point)

   Confounding- occurs when an extraneous variable correlates with both the disease and the exposure dependent variable in the health-related event under study (1 point)
2. Selection bias- the selection of individuals, groups or data for analysis that results in a non-random, biased sample that is not representative of the population from which it is taken (1 point)

   Information bias- bias that results from systematic error in the measurement of a variable (1 point)
3. (1 point for each correct answer)

   Bacterial

   Fungal

   Bacterial

   Bacterial

   Fungal

Islip Invitational - Repaired practice edition - Answer key

Division C | Disease Detectives | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

4. (6 points) Authored replacement for source item Question 4: Solve this case and explain your reasoning: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

reasoning: Selection is based on banquet attendance, not illness status.
Exposure and outcome are reconstructed for the full defined cohort.
Food-specific risks and RR can be estimated if ascertainment and denominators are adequate. (3 points).

conclusion: This is a retrospective cohort, despite collecting the information after the event. (3 points).

8. (6 points) Authored replacement for source item Question 8: Solve this case and explain your reasoning: Six cases have onset on day 1, eight on day 2, and two on day 3. Their report dates are all day 5.

reasoning: An onset curve has three bins with counts 6, 8, and 2.
A report-date curve would incorrectly look like 16 simultaneous reports if interpreted as onset.
Use onset timing to generate source hypotheses and assess completeness before claiming the source. (3 points).

conclusion: The time variable changes the meaning of the curve. (3 points).

repair-case. (4 points) Authored written substitute for unavailable source material: Why not compute risk from an ordinary case-control sample fraction?

Unit: Identifying study designs.
Worked-case data: Investigators interview every attendee on a banquet roster after an outbreak and compare illness by food consumed. Identify the design.

answer: Investigators choose case/control numbers; those fractions do not represent source-population disease occurrence. (4 points).